

SIMATS ENGINEERING

CO3 – ASSESSMENT TOOL 2

GAME BASED LEARNING

Course Name: Deep Learning Course Code: MLA04

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Question 01 – Game Based Learning Activity

The activity consists of two rounds, with five levels in each round. The objective is to understand CNN architecture, filters, parameter sharing, regularization, AlexNet, ResNet and real-world applications.

ROUND 1: CNN Architecture Builder

Level 1 – CNN Layer Matcher

Convolution extracts features using learnable filters. Pooling reduces spatial dimensions and computation. Flatten converts feature maps into a one-dimensional vector. Fully Connected layers use the extracted features for final classification.

Level 2 – Filter Finder

Filters/kernels detect features such as edges, textures and patterns. A 3×3 kernel is commonly used. Stride controls how far the kernel moves, while padding adds border values to control the output size.

Level 3 – Parameter Sharing Puzzle

In convolution, the same filter weights are reused at different image locations. This greatly reduces the number of parameters compared with a fully connected layer. For one filter, $\text{parameters} = \text{kernel height} \times \text{kernel width} \times \text{input channels} + 1 \text{ bias}$.

Level 4 – CNN Architecture Debugger

A valid CNN should maintain compatible dimensions between layers. A common sequence is Convolution → Activation → Pooling → Convolution → Pooling → Flatten → Fully Connected. Incorrect layer order or incompatible dimensions must be corrected before classification.

Level 5 – AlexNet vs ResNet Challenge

AlexNet is an early deep CNN that demonstrated strong image-classification performance using convolutional layers, pooling and fully connected layers. ResNet introduces residual/skip connections, allowing information and gradients to pass through deeper networks and helping reduce the degradation problem.

ROUND 2: CNN Feature Detective

Level 1 – Edge Detection Explorer

A 3×3 edge-detection filter can emphasize sudden changes in pixel intensity. During convolution, the kernel slides across the image and produces a feature map where strong responses indicate likely edges.

Level 2 – Feature Map Detective

Early CNN layers usually learn simple features such as edges and corners. Middle layers combine these into textures and shapes. Deeper layers represent more complex patterns and object-level features.

Level 3 – Regularization Rescuer

Overfitting occurs when a CNN performs very well on training data but poorly on unseen data. Suitable methods include dropout, data augmentation, weight regularization and early stopping. These improve generalization.

Level 4 – CNN Architecture Race

For a relatively simple image-classification task, AlexNet can be considered as a basic historical architecture. For deeper and more demanding image-classification problems, ResNet is generally preferable because residual connections make training deep networks easier.

Level 5 – CNN Application Master

Example: medical image analysis. A CNN can take a medical image as input, use convolution layers to extract visual features, pooling to reduce dimensions, and fully connected/output layers to classify the image. A ResNet-style architecture can be selected when deeper feature extraction is required. The model should be trained with suitable regularization and validated on unseen images.

Conclusion

The two-round game-based activity provides a structured way to learn CNN concepts. Round 1 focuses on building and understanding CNN architectures, while Round 2 focuses on feature extraction, regularization, architecture comparison and practical applications. Completing all ten levels demonstrates understanding of important CNN concepts and their use in real-world computer vision.

Student Declaration

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Signature of Student: _____ Date: _____